

ISM Note 11

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1. Riemann Problem

The equation we need to solve is the following:

$$\frac{\partial u}{\partial t} + \nabla \cdot F(u) = 0$$

And the initial condition is:

$$u(x, 0) = \begin{cases} u_L, & x < 0 \\ u_R, & x \geq 0 \end{cases}$$

This problem is important for astrophysical simulations, since we need to discretize the continuous fluid into finite volumes, and **the Riemann problem occurs at every interface between two adjacent cells**.

The problem we need to solve here is called the **sod shock tube problem**, which is a special case of the Riemann problem. The solution to this problem consists of three waves: a **shock wave**, a **contact discontinuity**, and a **rarefaction wave**. The shock wave is a discontinuity in the fluid properties that propagates through the medium, while the contact discontinuity is a surface across which the fluid properties are continuous but the density is not. The rarefaction wave is a smooth expansion wave that propagates through the medium. One sketch of the solution is shown in the figure below:

Here $n = k + m$ is the **governing parameter**, and k is the number of independent dimensions. Now we introduce the dimensionless parameters, which are given by

$$\Pi = \frac{a}{a_1^p a_2^q \dots a_k^r}$$

$$\Pi_i = \frac{b_i}{a_1^{p_i} a_2^{q_i} \dots a_k^{r_i}}$$

If we plug these dimensionless parameters into the original function, we can rewrite the original function as

$$\Pi = \frac{1}{a_1^p a_2^q \dots a_k^r} f(a_1, a_2, \dots, a_k, \Pi_1 a_1^{p_1} a_2^{q_1} \dots a_k^{r_1}, \Pi_2 a_1^{p_2} a_2^{q_2} \dots a_k^{r_2}, \dots, \Pi_m a_1^{p_m} a_2^{q_m} \dots a_k^{r_m})$$

And we can define a new function to simplify the above equation, which is given by

$$\Pi = F(a_1, a_2, \dots, a_k, \Pi_1, \Pi_2, \dots, \Pi_m)$$

The key thing is, for any a_i with independent dimensions, changign it by an arbitrary factor λ , we can always pass in a unit system to keep $a_2 \dots a_k$ unchanged. This means that **the physics does not depend on the choice of the unit system**. Therefore this implies that F is independent of a_i ! So we can drop out all the a_i in the above equation, and we can get

$$\Pi = \Phi(\Pi_1, \Pi_2, \dots, \Pi_m)$$

By doing the above analysis, we decrease the dimensions from n to $m = n - k$.

Therefore we can rewrite the original function as

$$f(a_1, a_2, \dots, a_k, b_1, b_2, \dots, b_m) = a_1^p a_2^q \dots a_k^r \Phi(\Pi_1, \Pi_2, \dots, \Pi_m)$$

This is the so-called **Buckingham Pi theorem**, which is a key result in dimensional analysis.

Example1: Hydrodynamic Drag in a Long Pipe

* hydrodynamic drag in a long pipe

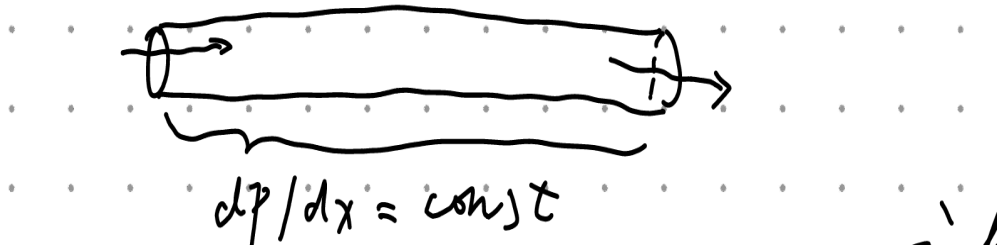


Fig 2. Hydrodynamic Drag in a Long Pipe

As an example, we can apply the dimensional analysis to the problem of hydrodynamic drag in a long pipe. We need to solve for the pressure gradient across the cross section of the pipe $\frac{dP}{dx}$, we can write it like

$$\frac{dP}{dx} = f(\rho, \mu, D, u)$$

where ρ is the fluid density, μ is the dynamic viscosity, D is the pipe diameter, and u is the flow velocity.

Note that the independent variables here are only 3, without loss of generality, we can choose them as u , D , and ρ , then we can get

$$\left[\frac{dP}{dx}\right] = [u]^2 [D]^{-1} [\rho]$$

$$[\mu] = [u] [D] [\rho]$$

Then we can write down the dimensionless parameters, which are given by

$$\Pi = \frac{\frac{dP}{dx}}{u^2 D^{-1} \rho}$$

$$\Pi_1 = \frac{\mu}{uD\rho}$$

According to the Buckingham Pi theorem, we can rewrite the original function as

$$\frac{dP}{dx} = (u^2 D^{-1} \rho) \Phi\left(\frac{\mu}{uD\rho}\right)$$

So with dimensional analysis, we simplify the original function and get the solution in a more compact form. The experiment gives the relation plot between Π and Π_1^{-1} , which is shown in the figure below:

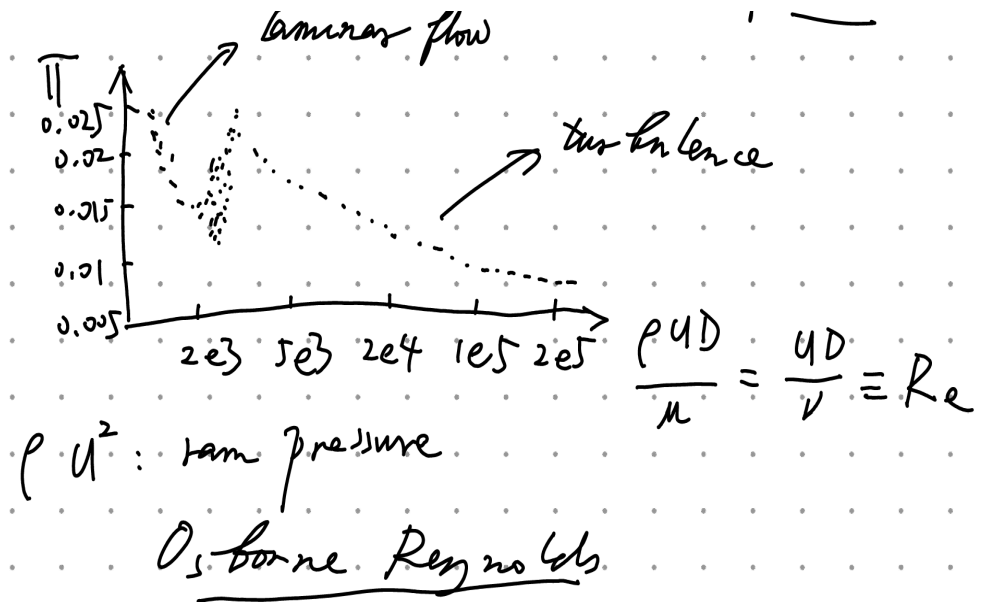


Fig 2. The relation between the drag coefficient and the Reynolds number.

The left smooth curve represents the **laminar flow**, and the right smooth curve represents the **turbulent flow**. The transition between these two curves is quite random and is called the **transition flow**. The inverse of Π_1 is called the **Reynolds number**, which is given by

$$Re = \frac{uD\rho}{\mu}$$

And we can find that the transition from laminar flow to turbulent flow occurs at $Re \sim 10^3$.

Example2: The Turbulent Flow

Also consider the long pipe, but now with a turbulent flow. We impose a pressure difference P between the two ends of the pipe, and we wait for some time τ then we measure the volume of fluid Q dropped out from the pipe. The question is, how does P depend on these parameters? In the same analytic approach, we can write it like

$$P = f(\tau, Q, \mu, \rho)$$

The independent variables are only 3 as well, and we can choose them as τ , Q , and μ , then we can get

$$[P] = [\tau]^{-1}[Q]^0[\mu]$$

$$[\rho] = [\tau][Q]^{-\frac{2}{3}}[\mu]$$

Then we can get the dimensionless parameters, which are given by

$$\Pi = \frac{P}{\mu\tau^{-1}}$$

$$\Pi_1 = \frac{\rho}{\mu\tau Q^{-\frac{2}{3}}}$$

Then we can rewrite the original function as

$$P = \mu\tau^{-1}\Phi\left(\frac{\rho}{\mu\tau Q^{-\frac{2}{3}}}\right)$$

Example3: Taylor-Sedov blast wave

Assume ambient pressure is negligible, and we want to find the radius of the blast wave as a function of time. First we need to write down the governing parameters, which are given by t , E , and ρ . Then we can get the dimensionless parameter, which is given by

$$\Pi = \frac{R}{(Et^2/\rho)^{1/5}}$$

After that, we can rewrite the original function as

$$R = (Et^2/\rho)^{1/5} \cdot \text{const}$$

Sedov is the energy conserving phase of SNR. The system is self-similar, which means the solution at different time is the same, just with a different scale.

Supernova Remnant (SNR)

Sedov Phase

The first is the Sedov phase, which is the **energy conserving phase**. For this phase, we have

$$t_{\text{dyn}} \ll t_{\text{cool}}$$

so the cooling is negligible. The energy can be given by

$$e = \frac{1}{2}v^2 = \frac{9}{32}\dot{R}^2$$

where v is the velocity of the "swept-up" gas by shock in lab frame , and the coefficient $\frac{9}{32}$ comes from the shock jump condition. Since the kinetic energy is comparable to the internal energy, so the total energy is given by

$$E_{\text{total}} = \frac{4\pi}{3}R^3\rho_0\left(e + \frac{1}{2}v^2\right) = \frac{4\pi}{3}\rho_0R^3\dot{R}^2 = E_{\text{SN}}$$

So we can solve for $R(t)$ from the equation above, then we get

$$R(t) = \left(\frac{25}{3\pi}\right)^{1/5} \left(\frac{E_{\text{SN}}t^2}{\rho_0}\right)^{1/5}$$

And

$$\dot{R}(t) = \frac{2}{5} \left(\frac{25}{3\pi}\right)^{1/5} \left(\frac{E_{\text{SN}}}{\rho_0 t^3}\right)^{1/5}$$

This agrees with the result we get from dimensional analysis above.

Snowplow Phase

The second is the snowplow phase, which is the **momentum conserving phase**. For this phase, we have

$$M_{\text{shell}}(t)\dot{R}(t) \sim R^3\rho_0\dot{R}(t) \sim R^3\frac{dR}{dt} = \text{const}$$

And then we can get

$$R(t) \propto t^{1/4}$$

$$v(t) \propto t^{-3/4}$$

The details of this will be discussed in the next lecture.